Rahul Juluru

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EXPERIENCE

University of Oklahoma **Aug 2024 - Present**

Software Engineer - Backend Engineering, Distributed Systems, ML Infrastructure *Norman, OK*

- Architected and deployed a containerized GCP data pipeline (Cloud Run + Pub/Sub) that automated a 5-stage manual preprocessing workflow for 1.8 TB of research data, reducing end-to-end processing time from 12 hours to 45 minutes and eliminating an estimated 15 hrs/week of manual effort.
- Reduced asynchronous task failure rate from ~35% to under 5% by introducing Pydantic schema validation, structured JSON logging, and exponential-backoff retry orchestration across a GCP Dataflow pipeline, enabling reliable processing of 12,000+ documents/day.
- Boosted feature extraction throughput by 4x using Python multiprocessing and shm-based shared memory pools to parallelize vector computation across 32-core GCP Compute Engine instances, eliminating I/O bottlenecks for downstream ML training.
- Instrumented end-to-end pipeline observability using structured JSON logging, GCP Cloud Monitoring dashboards, and alerting policies - reducing mean time to detect (MTTD) silent failures from hours to under 10 minutes and enabling proactive incident response across all 5 pipeline stages.
- Authored a modular integration test suite covering critical pipeline stages (ingest, transform, export), catching schema regressions and data contract violations pre-deployment - reducing production bug discovery rate by ~70% across iterative research data releases.

Neospark **Jun 2022 - Jun 2024**

Software Engineer *Hyderabad, India*

- Shipped a self-serve internal reporting portal (Node.js / Express / React) using RESTful API design that replaced ad-hoc spreadsheet workflows across 5+ teams - reclaiming 8 hrs/week of analyst time and eliminating a 2-day manual reporting cycle.
- Accelerated multi-report dashboard load time by 25% (1.8s → 1.35s, within the 1.5s SLA target) by batching asynchronous API calls and eliminating redundant sequential fetches in the Express response layer.
- Refactored frontend state management from prop-drilling to Redux and Context API, reducing wasted renders from 340ms to 95ms per interaction (profiled via React DevTools) and raising Lighthouse performance score from 61 to 84.
- Established a JWT-based authentication system with role-based access control, supporting 3 user permission tiers across the internal portal - eliminating shared credential usage and enabling per-team audit trails for compliance reporting.
- Owned end-to-end PostgreSQL schema design and migration strategy for the reporting portal, authoring versioned migrations using a structured rollback plan - reducing schema-related deployment incidents to zero across 6+ production releases.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, JavaScript, TypeScript, SQL, HTML/CSS, Go
- **Frameworks & Tools:** React.js, Next.js, Node.js, Express.js, FastAPI, Socket.io, Kubernetes, Docker, AWS, GCP, Redis, PostgreSQL, GitHub Actions

SOFTWARE PROJECTS

Smart Expense Tracker - AI-Powered Expense Management System **Aug 2025 - Dec 2025**

- Next.js, Express.js, FastAPI, GPT-4o, Tesseract, OpenCV, Redis, PostgreSQL, Docker, k6
- Structured a fault-tolerant microservices architecture (Next.js client, Express API gateway, FastAPI ML service) deployed onto containerized infrastructure, ensuring isolation and independent scaling of compute-heavy OCR workloads.
- Built a receipt classification pipeline combining Tesseract OCR + OpenCV preprocessing with zero-shot GPT-4o prompting, evaluated on a manually labeled set of 850+ receipts across 15 categories, achieving 92% top-1 accuracy vs. 61% for a regex-based baseline.
- Tuned PostgreSQL query latency and introduced a Redis read-through caching layer; validated system resilience via k6 load testing, sustaining 600 req/sec with a P95 latency of 120ms without degradation.

DistriLimit - Distributed Rate Limiting System **Apr 2025 - Jul 2025**

- Redis, Lua, Python, Docker, Token Bucket Algorithm, Sliding Window
- Developed a distributed rate limiter using the Token Bucket algorithm and a sliding-window strategy, utilizing Redis as a centralized atomic counter to enforce global quotas across 10+ simulated microservices nodes.
- Analyzed and resolved double-counting race conditions using Lua scripts for atomic Redis updates, reducing synchronization overhead and maintaining <5ms latency overhead per request under 2,000 req/sec load.
- Benchmarked fault tolerance by simulating Redis node failures, implementing a local fallback strategy that ensured system availability with 99.9% uptime during high-traffic simulated DDoS events.

CollabSpace - Real-Time Collaborative Workspace **Nov 2024 - Feb 2025**

- Next.js, Socket.io, Redis Pub/Sub, PostgreSQL, Tiptap, JWT, Docker, GitHub Actions
- Engineered a real-time collaborative document editor supporting concurrent multi-user editing via Socket.io and operational-transform conflict resolution - maintaining document consistency across 50+ simultaneous sessions in load tests.
- Enforced role-based access control (RBAC) with JWT authentication across 3 permission tiers (owner, editor, viewer), enabling per-document sharing without exposing global user data - zero privilege-escalation incidents across security test suite.
- Streamlined live presence sync by routing user-state events through Redis Pub/Sub, decoupling WebSocket instances and achieving sub-100ms latency for cursor updates across distributed server nodes.

EDUCATION

University of Oklahoma **Aug 2024 - May 2026**

Master of Science, Computer Science - 3.78/4.0 *Norman, OK*

- **Coursework:** Distributed Systems, Advanced Operating Systems, Cloud Computing, ML Infrastructure, Data Structures, Algorithms, DBMS, Web Development, Computer Networks

CVR College of Engineering **Aug 2020 - May 2024**

Bachelor of Technology, Information Technology - 9.35/10.0 *Hyderabad, India*